

retrospective design (e.g. withdrawals from the study, proportion of participants agreeing to take part in the study) nor for studies of a single-group design (confounders). Also in the context of surgical studies for epilepsy, where blinding of participants and outcome assessors is often impossible and only objective, non-validated methods for collection of seizure outcome data exist, the global quality rating of the study was influenced by two components (D and E) for most studies. Only five of the RCTs included in this review attempted blinding; therefore we automatically assigned the judgement for 175 of 182 studies (97%) as 'weak' according to the definition provided in the tool. On the basis of global rating, this meant that we could judge 97% at the most to be of 'moderate' quality due to lack of blinding alone. Then the difference between a 'moderate' and a 'weak' global rating was dictated by component E ('strong' or 'weak'). Essentially, the global rating reflects the quality of data collection methods rather than overall quality.

Furthermore, for the component "B. Study design", we made an assumption that the study design of 144 'retrospective case series' (one group before and after intervention) not specifically referred to in the EPHPP tool was of 'moderate' quality - the same quality assigned to a prospective cohort study. We made this judgement to avoid the design of the study alone dictating the overall quality rating (i.e. all retrospective designs are 'weak') to separate out the 144 studies based on other quality criteria. If instead, more fitting with National Institute for Health and Care Excellence (NICE) guidelines (Mann 1996), we had assumed all retrospective designs to be of 'weak' quality, the global rating would have been as follows: 'strong': five studies; 'moderate': six studies; and 'weak': 171 studies. Such ratings would not have provided us with any useful information regarding the relative quality of included studies.

In hindsight, given the context of surgery for epilepsy and our prior knowledge of the likely design of included studies based on the Tonini 2004 review, it would have been more appropriate for us to design our own quality assessment tool for the review based on what we know to be clinically important in studies of surgical interventions. Given our lack of confidence in the global ratings assigned to included studies by the EPHPP tool, we believe it would be inappropriate to conduct sensitivity analyses based on the global quality assessments (see [Effects of interventions](#) for subgroup and sensitivity analyses performed).

Assessment of risk of bias in studies reporting multi-variable prognostic models according to the Quality in Prognostic Studies (QUIPS) tool

Twenty-eight studies reported a multi-variable prognostic model including one or more of the factors of interest to us (Althausen 2013; Boesebeck 2007; Cossu 2005; Cossu 2008; Elsharkawy 2008a; Elsharkawy 2009a; Gelinis 2011; Janszky 2003a; Jennum 1993; Kim 2009; Kim 2010a; Lopez-Gonzalez 2012; Madhavan 2007; McIntosh 2012; O'Brien 2000; Paolicchi 2000; Phi 2009; Radhakrishnan 1998; Rossi 1994; Sagher 2012; Sarkis 2012; Schramm 2011; Spencer 2005; Tezer 2008; Theodore 2012; Walz 2003; Wyler 1995; Yang 2011) (see "Multi-variable analyses" in [Effects of interventions](#) below for additional details of these results and reported results).

The QUIPS tool considers six domains (see [Table 5](#) for judgements for each domain for each study).

1. Study participation: judge risk of selection bias (likelihood that the relationship between prognostic factor (PF) and outcome is different for participants and eligible non-participants)

We judged 11 of 28 studies to be at 'low' risk of selection bias; the population of interest and the method of sample recruitment are well described and the samples seem to match the characteristics of the source population. We judged 15 of the 28 studies to be at 'moderate' risk of selection bias due to uncertainties or limited information regarding the population of interest or the method of sample recruitment, or both, to enable judgement on whether the sample matched the source population. We judged two studies to be at 'high' risk of selection bias: one due to a selective sample that is unlikely to represent the source population (Janszky 2003a), and one due to very limited information regarding the population of interest, the method of sample recruitment, and the characteristics of the sample (Rossi 1994).

2. Study attrition: judge risk of attrition bias (likelihood that the relationship between PF and outcome is different for completing and non-completing participants)

Twenty-two of the 28 studies were of a retrospective design, so this domain was not applicable. Six studies were of a prospective design. We judged that two of these studies were at 'low' risk of attrition bias, as intention-to-treat analyses were planned in the case of withdrawals or losses to follow-up, so all participants contributed to outcome assessment (Schramm 2011; Wyler 1995). We judged three to be at 'moderate' risk of attrition bias due to lack of information reported about withdrawals and losses to follow-up (Spencer 2005; Theodore 2012; Walz 2003), and one study to be at 'high' risk of attrition bias due to exclusion of participants with missing data and uncertainty over whether participants were recruited prospectively or retrospectively (Radhakrishnan 1998).

3. Prognostic factor measurement: judge risk of measurement bias related to how PF was measured (differential measurement of PF related to level of outcome)

We judged 22 of the 28 studies to be at 'moderate' risk of measurement bias due to unclear definitions of prognostic factors and limited information regarding how data were collected. Four studies judged to be at 'low' risk of measurement bias provided detailed definitions of prognostic factors and methods of measurement and data collection (Cossu 2008; McIntosh 2012; Sagher 2012; Sarkis 2012). We judged two studies to be at 'high' risk of measurement bias; in one multi-centre study, it is likely that researchers collected data using different methods across centres (Madhavan 2007), and in the other study, a large proportion of prognostic factor data was missing, which is likely to have had an impact on analyses (Rossi 1994).

4. Outcome measurement: judge risk of bias related to measurement of outcome (differential measurement of outcome related to baseline level of PF)

Recurrence of seizures in outpatients is generally patient-reported and therefore difficult to measure in a valid and reliable way; validated scales such as the Engel Class Scale as described above exist for assessment of post-surgical outcome.

We judged 18 of 28 studies to be at 'moderate' risk of bias; it is unclear exactly how and/or when outcome data had been collected

and/or how outcome was defined. We judged two studies to be at 'high' risk of bias: in [Althausen 2013](#), researchers measured outcome in variable ways (taken only from patient reports for some participants, and using supplementary data from medical records for other participants); [Kim 2009](#) did not measure the outcome according to a known scale such as the Engel Class Scale and provided no information on when study authors recorded outcome. We judged the remaining eight studies to be at 'low' risk of bias: investigators reported clear information about how they collected outcome data; they measured outcome according to a known scale such as the Engel Class Scale; and they recorded this information at the same time for all participants.

5. Study confounding: judge risk of bias due to confounding (i.e. the effect of PF is distorted by another factor that is related to PF and outcome)

Twenty of the 28 studies were of a single-group design, so this domain was not applicable. We judged one RCT, [Wyler 1995](#), to be at 'low' risk of bias due to confounding, as the randomised design should remove confounding and it is confirmed in the study that groups were balanced at baseline. We judged seven studies to be at 'moderate' risk of bias of confounding: one study of an RCT design did not demonstrate whether groups were balanced at baseline and which variables of interest were prognostic factors and which were confounders ([Schramm 2011](#)); another study made reference to confounders and interactions in a generalised estimating equations model but did not specify which variables of interest were prognostic factors and which were confounders ([Sagher 2012](#)); and the remaining five studies made reference to "confounders" but did not adequately define the variables and/or did not specify which variables of interest were prognostic factors and which were confounders ([Althausen 2013](#); [Cossu 2008](#); [Elsharkawy 2008a](#); [Gelinis 2011](#); [Janszky 2003a](#)).

6. Statistical analysis and reporting: judge risk of bias related to statistical analysis and presentation of results

We judged only one study to be at 'low' risk of bias because researchers described statistical analysis well, performed modelling based on clinical relevance, and did not selectively report results ([O'Brien 2000](#)). We judged the other 27 studies to be at 'moderate' or 'high' risk of bias (13 'moderate' and 14 'high') due to use of unclear or inappropriate statistical methods, selection of variables based on statistical significance, and selective reporting of results. We have further discussed these issues and their likely impact on the analyses under "Multi-variable analyses" in [Effects of interventions](#) (below).

Overall, we judged one study to be at 'high' risk of bias in three domains ([Rossi 1994](#)), three studies to be at 'high' risk of bias in two domains ([Althausen 2013](#); [Janszky 2003a](#); [Madhavan 2007](#)), and 11 studies to be at 'high' risk of bias in one domain ([Cossu 2008](#); [Elsharkawy 2008a](#); [Elsharkawy 2009a](#); [Jennum 1993](#); [Kim 2009](#); [Paolicchi 2000](#); [Radhakrishnan 1998](#); [Sarkis 2012](#); [Spencer 2005](#); [Tezer 2008](#); [Theodore 2012](#)). The remaining 13 studies were not judged to be at 'high' risk of bias in any domain ([Boesebeck 2007](#); [Cossu 2005](#); [Gelinis 2011](#); [Kim 2010a](#); [Lopez-Gonzalez 2012](#); [McIntosh 2012](#); [O'Brien 2000](#); [Phi 2009](#); [Sagher 2012](#); [Schramm 2011](#); [Walz 2003](#); [Wyler 1995](#); [Yang 2011](#)).

Effects of interventions

See: [Summary of findings for the main comparison](#); [Summary of findings 2](#)

Overall outcome of surgery compared to medical treatment

Two studies randomised participants to a surgical or control (medical) intervention ([Dwivedi 2017](#); [Wiebe 2001](#)).

[Dwivedi 2017](#) randomised 116 participants aged 18 years or younger to receive either appropriate brain surgery or continuing medical therapy.

All 57 allocated to the surgical intervention completed 12-month follow-up. One of the 59 allocated to continuing medical treatment was lost to follow-up. Researchers used an intention-to-treat approach to analysis and analysed all randomised participants in the allocated groups, regardless of the intervention received. Serious adverse events occurred in 19 participants (33%) in the surgery group and in none in the medical therapy group. These events included monoparesis in two participants (following temporal lobectomy or resection of parietal focal cortical dysplasia); hemiparesis in 15 (following hemispherotomy); and generalized hypotonia and language deficits in one (following frontal lobectomy). Study authors provided details on outcome at 12-month follow-up. None of these events appears to have been transient.

[Wiebe 2001](#) randomised 40 participants to each intervention group and followed them up for 12 months for assessment of outcome.

Four participants allocated to the surgery group did not undergo surgical intervention (one declined surgery, two were deemed not eligible for surgery based on pre-operative testing, and one did not have seizures during pre-operative testing). Researchers took an intention-to-treat approach to analysis and analysed all randomised participants in the allocated groups, regardless of the intervention received.

Four participants experienced adverse events from surgery: one had a small thalamic infarct causing sensory abnormalities in the thigh, one had a wound infection, and two had decline in verbal memory that interfered with their occupations at one year. In addition, 22 participants in the surgery group experienced asymptomatic, superior sub-quadrantic visual field defects; seven experienced depression; and one developed transient psychosis. The only adverse event reported in the medical treatment group was depression (eight participants).

Primary outcome: seizure outcome

(See also [Summary of findings for the main comparison](#).)

At one year, in [Wiebe 2001](#), 23 of 40 (58%) participants in the surgery group were free from seizures impairing awareness compared to 3 of 40 (8%) in the medical treatment group (risk ratio (RR) 7.67, 95% confidence interval (CI) 2.50 to 23.51; $P = 0.0004$; [Analysis 1.1](#)), and in [Dwivedi 2017](#), 44 of 57 (77%) participants in the surgery group were free from seizures (ILAE Class 1) compared to 4 of 59 (7%) in the medical treatment group (RR 11.33, 95% CI 4.37 to 29.64; $P < 0.00001$; [Analysis 1.1](#)). Pooling of data from the two studies showed that the RR of seizure freedom in the surgery group compared to the medical group was 9.78 (95% CI 4.73 to 20.21; $P < 0.00001$;

Table 5. Risk of bias in prognostic studies according to QUIPS tool

Author	1. Study participation		2. Study attrition		3. Prognostic factor measurement		4. Outcome measurement		5. Study confounding		6. Statistical analysis and reporting	
	Judgement	Risk of bias	Judgement	Risk of bias	Judgement	Risk of bias	Judgement	Risk of bias	Judgement	Risk of bias	Judgement	Risk of bias
Althausen 2013	Population of interest well described and sample recruited seems to match this; however sample restricted to those who returned post-operative questionnaires; may exclude some of those with poorer intellectual/seizure outcomes?	moderate	Retrospective, so NA	NA	Most variables well defined, some not; unclear how data were collected and whether data were complete	moderate	Outcome patient-reported, therefore likely to be highly subjective; also unclear when outcome was measured and if additional medical records were used for some/all participants	high	Most variables well defined, some not; unclear how data were collected and whether data were complete; some adjustment seems to have been done but not for all relevant variables	moderate	Insufficient information provided in the methods; some variables included in the model without pre-specification and no definitions; results selectively reported	high
Boesebeck 2007	Population of interest well described and sample seems	moderate	Retrospective	NA	Prognostic factors entering the model well described, but not	moderate	Difficult outcome to measure	moderate	NA for studies of	NA	Model well described and based on clinical	moderate

Table 5. Risk of bias in prognostic studies according to QUIPS tool (Continued)

	to match this, but unclear on recruitment methods and inclusion criteria		tive, so NA	much information given on how these data were collected		in a valid and reliable way, but outcome definition is clear and the same for all participants; method of collecting outcome data not specified		a single-group design		value rather than statistical significance; however not all P values reported and regression coefficients reported rather than meaningful effect sizes		
Cossu 2008	Population of interest well described; sample well described; sample representative of the source population	low	Retrospective, so NA	NA	All variables well described; measurement techniques specified and patients all came from a single centre, so likely followed the same protocol	low	Difficult outcome to measure in a valid and reliable way; measured on a known scale; last follow-up meaning variability across measurement times	moderate	Age at surgery and pathology seem to have been considered as confounders; unclear what are confounders and what are prognostic factors	moderate	Analysis described clearly, but variables in the model based on statistical significance; results presented only for statistically significant variables	high
Cossu 2005	Population of interest well described and original sample matches this population; however	moderate	Retrospective, so NA	NA	Variables well defined, but very little information provided on how data were collected and analysed	moderate	Difficult outcome to measure in a valid and reliable way;	moderate	NA for studies of a sin-	NA	Methods well described on the model and variables, but reporting of results	moderate

Table 5. Risk of bias in prognostic studies according to QUIPS tool *(Continued)*

	er not all from original sample included in final analyses, and characteristics not described for the subgroup of interest who did undergo surgery		so NA				information provided on outcome classification, but not on when outcome was measured and how outcome was measured		gle-group design		highly selective, with no useful information provided at all	
Elsharkawy 2008a	Population of interest and sample well described; seem to match	low	Retrospective, so NA	NA	All variables well defined, pre-operative workup seems the same but data collected retrospectively in a variety of ways, which could have led to variability across participants	moderate	Difficult variable to measure in a valid and reliable way; outcome well defined overall, but unclear at which time point modelling was conducted and potential variability in data collection across participants	moderate	Although all results presented according to "Group", differences between these groups do not seem to be considered; unclear if any variables are being treated as confounders or as prognostic	moderate	Model appropriate for the data, but variables included based on statistical significance and results selectively reported	high

Table 5. Risk of bias in prognostic studies according to QUIPS tool *(Continued)*

Risk of bias in prognostic studies according to QUADAS tool (continued)												
Study	Population	Low	Retrospective	NA	Variables	Moderate	Outcome	Low	Prognostic variables	NA	Model	High
Elsharkawy 2009a	Population of interest well defined, and sample recruited seems to match this; very clear how sample was recruited; large sample size	low	Retrospective, so NA	NA	Most variables well defined, and data recorded and collected following the same protocol; however definitions of some variables unclear due to variable terminology, which makes it unclear which variables have been analysed	moderate	Difficult outcome to measure in a valid and reliable way, but methods seem satisfactory (same time points, single protocol, multiple data sources)	low	NA for studies of a single-group design	NA	Model appropriate for the data but variables included based on statistical significance; some variable definitions unclear and results selectively reported	high
Gelinas 2011	Population of interest and sample recruited well described and seem compatible; uncertainty only around the definition of "paediatric"	moderate	Retrospective, so NA	NA	All variables well defined and method of measurement seems valid and consistent for most; some variability possible for one of the most important variables in the study - subjective choice	moderate	Difficult outcome to measure in a valid and reliable way; measured at a single time point for analysis and judged by an expert, so seems reasonable	low	Potential "confounder"; type of surgery well defined but choice was subjective and likely to vary across participants; differences between groups considered,	moderate	Results well reported, but unclear exactly how many variables went into the model	moderate

Table 5. Risk of bias in prognostic studies according to QUIPS tool *(Continued)*

Risk of bias in prognostic studies according to QUADAS tool (continued)												
												and this variable included in regression analysis
Jan-szky 2003a	Population of interest well described; however given the clinical question of interest, the sample seems quite selective	high	Retrospective, so NA	NA	Prognostic factors well defined including definitions of key variables (IEDs, CFCs); unclear whether participants were submitted to 1 or 2 centres, so unclear if data collection methods are the same	moderate	Difficult outcome to measure in a valid and reliable way; outcome recorded according to a known scale, but unclear when outcome was measured (possibly variable across participants); little detail given about how data were collected	moderate	All apparent relevant confounders examined, well defined, and adjusted for in analysis of prognostic factors, but unclear whether participants were submitted to 1 or 2 centres, so unclear if da-	moderate	Methods describe models quite well; unclear how it was decided which variables went into the prognostic model; results selectively reported	high

Table 5. Risk of bias in prognostic studies according to QUIPS tool *(Continued)*

												ta col- lection meth- ods are the same
Jen- num 1993	Population of interest well described, and sample seems to match this, but unclear on recruitment methods and inclusion criteria	mod- erate	Ret- ro- spec- tive, so NA	NA	Prognostic factors entering the model mostly well described (except for extent of resection), but not much information given on how these data were collected	mod- erate	Difficult outcome to measure in a valid and reliable way; however outcome data collected in the same way for all participants and measured at the same time	low	NA for studies of a single-group design	NA	Very little information given about the model; only significant P values reported	high
Kim 2009	Population of interest well described; sample well described; seem to match	low	Ret- ro- spec- tive, so NA	NA	Definitions and measurements of some factors unclear	mod- erate	Difficult outcome to measure in a valid and reliable way; not defined according to a scale such as Engel Class; no information on when outcome is measured	high	NA for studies of a single-group design	NA	Methods well described, but variables entered into the model based on statistical significance and results selectively reported	high
Kim 2010a	Population of interest and sample well defined, but unclear whether patients were recruited from a single centre or from	mod- erate	Ret- ro- spec- tive, so NA	NA	Most variables well described but technical; not clear if participants came from a single centre, so different protocols may have been followed	mod- erate	Seizure outcome defined, but not the time point measured; unclear how	mod- erate	NA for studies of a single-group design	NA	Unclear which variables went into the model; results selectively reported	mod- erate

Table 5. Risk of bias in prognostic studies according to QUIPS tool *(Continued)*

	multiple centres, and if the type of epilepsy in the sample matches the population						data were collected					
Lopez-Gonzalez 2012	Population of interest well defined, and sample recruited seems to match this; very clear how sample was recruited	low	Retrospective, so NA	NA	Variables well described; pre-operative evaluations carried out in standardised fashion, but unclear how retrospective analysis was carried out (patient records extracted, etc.)?	moderate	Outcome briefly defined but no further information provided; unclear how outcome data were collected; multiple time points for the analysis accounted for in the analysis	moderate	NA for studies of a single-group design	NA	Statistical methods well described; variables well defined (but based on statistical significance) and results clearly tabulated (unclear why RRs were reported)	moderate
Madhavan 2007	Descriptions of population of interest and sample recruited not quite the same; inclusion criteria brief	moderate	Retrospective, so NA	NA	Likely that measurement of variables was different across centres; unclear how some variables have been defined and choice for analysis	high	Seizure outcome defined according to a known scale; time point not stated and unclear how outcome data were collected	moderate	NA for studies of a single-group design	NA	Unclear what kind of statistical model had been fitted; only significant variables entered into the model	high
McIntosh 2012	Population of interest well described; study sample matches population for characteristics of interest; all included in analysis	low	Retrospective, so NA	NA	Prognostic variables well defined; a lot of effort made to accurately collect data for all participants; participants came from a single centre	low	Difficult outcome to measure in a valid and reliable way, but a lot of effort made to correctly	low	NA for studies of a single-group design	NA	Methods well described and results well reported; however, choice of variables and categories based on	moderate

Table 5. Risk of bias in prognostic studies according to QUIPS tool *(Continued)*

Rate of risk of bias in prognostic studies according to QUADAS tool (continued)												
						classify participants; outcome measured at 1 time point in primary analysis				data and statisti- cal significance		
O'Brien 2000	Population of in- terest well defined and sample recruit- ed seems to match this; very clear how sample was recruit- ed	low	Ret- ro- spec- tive, so NA	NA	Potential for misclas- sification due to the nature of the prog- nostic factors, but ef- forts made to reduce risk; definitions well described but unclear why some participants were not included in some analyses	mod- erate	Difficult out- come to measure in a valid and reliable way; measured according to a known scale; could have been different if a different time of mea- surement was chosen due to vari- ability of outcome	mod- erate	NA for stud- ies of a sin- gle-group design	NA	Statistical analy- sis described very well, with no se- lective reporting of results	low
Paolic- chi 2000	Population of inter- est clear and sam- ple seems to match this; however in- sufficient informa- tion given regard- ing methods of re- cruitment	mod- erate	Ret- ro- spec- tive, so NA	NA	All definitions of vari- ables seem clear; little to no variation in col- lection/measurement methods seems to have occurred across participants, but un- clear if participants came from a single centre	mod- erate	Difficult out- come to measure in a valid and reliable way; un- clear when the out- come was measured - probably variable across par- ticipants, who may have come from multi- ple centres	mod- erate	NA for stud- ies of a sin- gle-group design	NA	Unclear which variables have ac- tually gone into the model, what "univariate con- ditions" means, and whether re- sults are selec- tively reported	high

Table 5. Risk of bias in prognostic studies according to QUIPS tool (Continued)

Phi 2009	Population of interest well defined, and sample recruited seems to match this; very clear how sample was recruited	low	Ret- ro- spec- tive, so NA	NA	Most variables well defined; some uncertainty on method of measurement for some variables and methods of analysis	mod- erate	Difficult outcome to measure in a valid and reliable way; unclear when the outcome was measured - probably variable across participants	mod- erate	NA for studies of a single-group design	NA	Methods section seems clear on methods and factor list, but unclear from the results exactly what has gone into the model	mod- erate
Rad- hakr- ishnan 1998	Sample recruited well described and seems representative of the population of interest; however unclear whether participants were recruited prospectively or retrospectively, and how complete follow-up information was	mod- erate	Un- clear whether par- tic- i- pants were re- cruit- ed prospec- tive- ly or ret- ro- spec- tive- ly (or both). 15 par- tic- i- pants ex- clud- ed	high	Variables seem to be measured in the same way for all participants and defined clearly; some variables subjective and taken from participant surveys by definition, so could induce bias; quantity of missing data unclear, as is how this was handled in analysis	mod- erate	Difficult outcome to measure in a valid and reliable way; could have been different if a different time of measurement was chosen due to variability of outcome	mod- erate	NA for studies of a single-group design	NA	Appropriate model, well described; however, unclear which variables had been entered into the multi-variate model (based on univariate analyses?) and if all results have been reported	mod- erate

Table 5. Risk of bias in prognostic studies according to QUIPS tool *(Continued)*

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Table 5. Risk of bias in prognostic studies according to QUIPS tool *(Continued)*

				ed the study is NA; un- clear how com- plete fol- low-up in- for- ma- tion was, and if this im- pact- ed analy- sis								
Rossi 1994	Population of in- terest vague and unclear, so uncer- tain if the sample represents this; little information about how the sample was recruit- ed; analysis per- formed on a subset of this sample	high	Ret- ro- spec- tive, so NA	NA	Variables well defined and methods/setting described in some de- tail, but unclear how the retrospective re- view was carried out; large proportion of missing data likely to have impacted the generalisability of analyses	high	Difficult out- come to measure in a valid and reliable way; out- come mea- sured at the same time point by a well-defined scale, but no details provided re- garding da- ta collection	mod- erate	NA for stud- ies of a sin- gle-group design	NA	Some uncertainty over exactly what has been done in terms of mod- elling; variables entered based on statistical sig- nificance; only P values reported without interpre- tation	high
Sagher 2012	Population of inter- est well described;	low	Ret- ro-	NA	All variables well de- scribed; multiple study	low	Seizure out- come mea-	low	Vari- ables	mod- erate	Statistical meth- ods well de-	mod- erate

Table 5. Risk of bias in prognostic studies according to QUIPS tool *(Continued)*

	sample recruited well described; seem to match		spec- tive, so NA	authors classified some variables, and data likely to have been collected in the same way for participants from a single centre		sured according to a known scale at specific time points; model builds in all follow-up of participants and data collected in the same way; participants from a single centre		well defined and separated by the 2 types of surgery; differences between variables mostly not tested; type of surgery included in GEE model		scribed, but unclear which variables went into the model and how effect sizes should be interpreted		
Sarkis 2012	Population of interest well described; sample recruited well described; seem to match	low	Ret- ro- spec- tive, so NA	NA	All variables well described; data likely to have been collected in the same way, with participants from a single centre	low	Seizure outcome defined according to a known scale; not completely clear how outcome data were collected; statistical model will account for variable follow-up length	moderate	NA for studies of a single-group design	NA	Statistical methods well described; however variables selected based on statistical significance and results selectively reported	high

Table 5. Risk of bias in prognostic studies according to QUIPS tool (Continued)

Schramm 2011	Large sample recruited; seems representative of the population of interest; however 35% of potentially eligible individuals not included for varying reasons	moderate	All participants in the study contributed data from 1 year and were included in an intention-to-treat analysis; secondary analysis was planned in the case of dropouts	low	Clinical/demographic data seem to have been collected in a reliable and valid way under the same protocol for all patients; however unclear what are prognostic variables and what are confounding factors	moderate	Difficult outcome to measure in a reliable and valid way but seems satisfactory; measured by a known scale at a single time point in the same way for all participants	low	Clinical/demographic data seem to have been collected in a reliable and valid way under the same protocol for all participants; however unclear what are prognostic variables and what are confounding factors; "confounders" do not appear to be ac-	moderate	Unclear which variables have been included in the model, exactly what type of analysis has been performed, and whether all results have been reported	moderate
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Table 5. Risk of bias in prognostic studies according to QUIPS tool *(Continued)*

counted for in the design, and unclear what type of analysis (if any) was done												
Spencer 2005	Large sample size, which is representative of the population of interest, but unclear how many other eligible individuals declined to take part or were excluded from analyses	moderate	Not enough information in the publications on participants lost to follow-up and those included in the analyses	moderate	Complete data collected under the same protocol; some scope for variation between participants; not all PF definitions are completely clear	moderate	Difficult outcome to measure in a reliable and valid way; relies on patient reporting and does not use a known scale	moderate	NA for studies of a single-group design	NA	Unclear whether the analysis was appropriate, which variables were entered into the model, and whether results were selectively reported	high

Table 5. Risk of bias in prognostic studies according to QUIPS tool *(Continued)*

			un- clear									
Tezer 2008	Population of in- terest and sam- ple well described; sample seems to match population of interest	low	Ret- ro- spec- tive, so NA	NA	Most variables well defined; unclear how some variables de- fined and analysed; a lot of effort made to avoid misclassification in data collection; par- ticipants came from a single centre	mod- erate	Difficult out- come to measure in a valid and reliable way, but a lot of effort made to correct- ly classify participants according to a known scale; how- ever, out- come mea- sured at variable time points	mod- erate	NA for stud- ies of a sin- gle-group design	NA	Unclear which variables went into the mod- el, what the out- come variable was, and results were selectively reported	high
Theodore 2012	Study sample seems to be rep- resentative of the population of inter- est; however not clear on how this sample was recruit- ed, how many eligi- ble participants de- clined to take part, whether any with- drawals occurred, etc.	mod- erate	Pros- pective study with 41 in- clud- ed par- tic- i- pants; no fixed study length (du- ra- tion of fol- low-up spec-	mod- erate	Exact prognostic fac- tors of interest from the PET scans unclear	mod- erate	Difficult out- come to measure in a valid and reliable way; could have been differ- ent if a dif- ferent time of measure- ment was chosen due to variabil- ity of out- come; not measured according to a known scale	mod- erate	NA for stud- ies of a sin- gle-group design	NA	Unclear how vari- ables are being analysed in the model and re- sults selectively presented (no nu- merical results at all)	high

Table 5. Risk of bias in prognostic studies according to QUIPS tool *(Continued)*

			i- fied), so com- plet- ing the study is not ap- plic- a- ble; no dropouts/ with- drawals/loss- es to fol- low-up men- tioned, and un- clear if any oc- curred									
Walz 2003	Sample recruited well described and seems representative of the population of interest; however unclear whether participants were recruited prospectively or retrospectively	mod- erate	No loss- es to fol- low-up men- tioned; rea- sons for ex- clu- sion	mod- erate	Data seem to have been collected the same way for all participants, in a way that should minimise bias. Some variables by definition could be prone to recall/misclassification biases, but clear definitions given (except dichotomised variables)	mod- erate	Difficult outcome to measure in a valid and reliable way; could have been different if a different time of measurement was chosen due to variability of outcome;	mod- erate	NA for studies of a single-group design	NA	Statistical analysis appropriate and all results presented; however choice of variables for the multi-variate model based on univariate results	mod- erate

Table 5. Risk of bias in prognostic studies according to QUIPS tool *(Continued)*

State of risk of bias in prognostic studies according to QUADAS tool (continued)											
			from analy- sis giv- en; not clear if the study is of a prospec- tive or ret- ro- spec- tive de- sign, so dif- fi- cult to judge with- drawals and dropouts			all partici- pants from a single cen- tre, and out- come mea- sured ac- cording to a known scale					
Wyller 1995	Population of in- terest not well de- fined; introduc- tion refers more to surgical technique than to the popula- tion it is being ap- plied to	mod- erate	Out- low come as- sess- ment at 1 year record- ed for all par-	Variables entered into the logistic regression model well defined and seem to have been collected reliably, but unclear if all partici- pants were from the sample centre and were following the same protocol	mod- erate	Difficult out- come to measure in a valid and reliable way, but every effort seems to have been made to ver- ify seizure outcomes of partici-	low	Ran- domi- sation has bal- anced groups in terms of any con- found- ing fac-	low	No statistical analysis de- scribed in the methods, only in the results; unclear exact- ly what has been done and whether selec- tive reporting oc- curred	mod- erate

Table 5. Risk of bias in prognostic studies according to QUIPS tool *(Continued)*

				tic- i- pants; no dropouts nor miss- ing da- ta			pants, and all out- comes mea- sured at the same time		tors; there- fore lo- gistic regres- sion should allow for iden- tifica- tion of prog- nostic factors with- out con- found- ing			
Yang 2011	Study sample ap- peared to be rep- resentative of the population of in- terest, but informa- tion is limited	mod- erate	Ret- ro- spec- tive, so NA	NA	Definitions and mea- surements of some factors unclear	mod- erate	Difficult out- come to measure in a reliable and valid way but seems satis- factory and measured according to a known scale; vari- ability in fol- low-up time handled in analysis	low	NA for stud- ies of a sin- gle-group design	NA	Statistical meth- ods unclear	mod- erate

CFC: complex febrile convulsion.
GEE: generalised estimating equation.
IED: interictal epileptiform discharge.
NA: not applicable.
PET: positron emission tomography.
PF: prognostic factor.
QUIPS: Quality in Prognostic Studies.